

1. Consider using this encoder-decoder model for machine translation. 1 point



This model is a "conditional language model" in the sense that the encoder portion (shown in green) is modeling the probability of the input sentence x .

- ☐ True
☒ False

2. In beam search, if you decrease the beam width $\bar{B}$, which of the following would you expect to be true? Select all that apply. 1 point

- ☐ Beam search will converge after fewer steps.
☒ Beam search will generally find better solutions (i.e. do a better job maximizing $P(y|x)$).
☒ Beam search will use up more memory.
☐ Beam search will run more quickly.

3. True/False: In machine translation, if we carry out beam search using sentence normalization, the algorithm will tend to output overly short translations. 1 point

- ☐ False
☒ True

4. Suppose you are building a speech recognition system, which uses an RNN model to map from audio clip x to a text transcript y . Your algorithm uses beam search to try to find the value of y that maximizes $P(y|x)$. 1 point

On a dev set example, given an input audio clip, your algorithm outputs the transcript $\hat{y}$ = "I'm building an A Eye system in Silly con Valley.", whereas a human gives a much superior transcript y^* = "I'm building an AI system in Silicon Valley."

According to your model,

$$P(\hat{y}|x) = 1.95 \times 10^{-7}$$

$$P(y^*|x) = 3.42 \times 10^{-9}$$

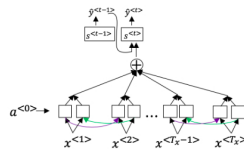
True/False: Trying a different network architecture could help correct this example.

- ☐ False
☒ True

5. Continuing the example from Q4, suppose you work on your algorithm for a few more weeks, and now find that for the vast majority of examples on which your algorithm makes a mistake, $P(y^*|x) > P(\hat{y}|x)$. This suggests you should not focus your attention on improving the search algorithm. 1 point

- ☐ True
☒ False

6. Consider the attention model for machine translation. 1 point



Further, here is the formula for $\alpha^{<t,t'>}$.

$$\alpha^{<t,t'>} = \frac{\exp(e^{<t,t'>})}{\sum_{t'=1}^T \exp(e^{<t,t'>})}$$

Which of the following statements about $\alpha^{<t,t'>}$ are true? Check all that apply.

- ☐ $\sum_{t'} \alpha^{<t,t'>} = 0$
☐ $\sum_{t'} \alpha^{<t,t'>} = -1$
☒ $\alpha^{<t,t'>}$ is equal to the amount of attention $y^{<t'>}$ should pay to $a^{<t'>}$
☐ We expect $\alpha^{<t,t'>}$ to be generally larger for values of $a^{<t'>}$ that are highly relevant to the value the network should output for $y^{<t'>}$. (Note the indices in the superscripts.)

7. The network learns where to "pay attention" by learning the values $e^{<t,t'>}$, which are computed using a small neural network: 1 point

We can replace $s^{<t-1>}$ with $s^{<t'>}$ as an input to this neural network because $s^{<t'>}$ is independent of $\alpha^{<t,t'>}$ and $e^{<t,t'>}$.

- ☐ True
☒ False

8. The attention model performs the same as the encoder-decoder model, no matter the sentence length. 1 point

- ☒ False
☐ True

9. Under the CTC model, identical repeated characters not separated by the "blank" character () are collapsed. Under the CTC model, what does the following string collapse to? 1 point

aaa_aaaaaa_rr_dddddddd_v_aaaaa_rrrr_kk

- ☐ aa rd var k
☐ ardvark
☐ aaaaaaaaaardddddddvaaaaarrkk
☒ aardvark

10. In trigger word detection, $x^{<t'>}$ is: 1 point

- ☐ Whether someone has just finished saying the trigger word at time t .
☒ Features of the audio (such as spectrogram features) at time t .
☐ Whether the trigger word is being said at time t .
☐ The t -th input word, represented as either a one-hot vector or a word embedding.

